

Gene Ontology and Functional Enrichments

- Homework Assignment #1 is due next Thur (1/29) at 11:59pm by online Blackboard submission (submit as .qmd file)
- Open RStudio
- Download from OneDrive the 02_Gene_Ontology.qmd file
- run the first two code chunks (through installing packages):

The general idea behind functional enrichments

- Genomic analyses frequently lead to lists with hundreds of genes:
 - differential expression analysis (set of genes up/down regulated)
 - genes present in one organism and absent in another
 - genes with a particular transcription factor binding site in their promoters (or other regulatory elements)
 - genes that have phenotypic difference from wild type in a high-throughput screen
 - genes with certain physical properties (found on a particular chromosome, have/do not have introns, found near telomeres, are longer or shorter, have higher or lower GC content...)
- It's impossible to make sense of a list of hundreds of genes one by one
- It's much easier to make sense of a large dataset by identifying over-represented shared functions within a set of genes

Gene Ontology (GO): A common language to describe gene products

- The Gene Ontology Consortium began in 1998 as a collaboration among model organism databases (FlyBase, Saccharomyces Genome Database (SGD), Mouse (MGD))
- Now consists of those original members, plus other model organism database members and software developers
- An ontology is a formal representation of concepts and the relationships among them
- The idea behind a gene ontology is to formally and consistently define (annotate) gene products within and between organisms
- The original goal was to provide consistent annotation of gene products within and between organisms

Gene Ontology (GO)

More specifically, the goals of the GO consortium were to:

- 1) Develop and maintain a controlled vocabulary for gene product attributes
- 2) Continuously annotate genes and gene products
- 3) Provide a database to access annotation data, and tools to enable functional interpretation (e.g., gene set enrichment analysis)

<http://geneontology.org/>

GO Terms

There are 3 gene ontologies (classifications):

Biological Process: 25,153 terms

The larger processes, or 'biological programs' accomplished by multiple molecular activities. Examples of broad biological process terms are [DNA repair](#) or [signal transduction](#).

Molecular Function: 10,143 terms

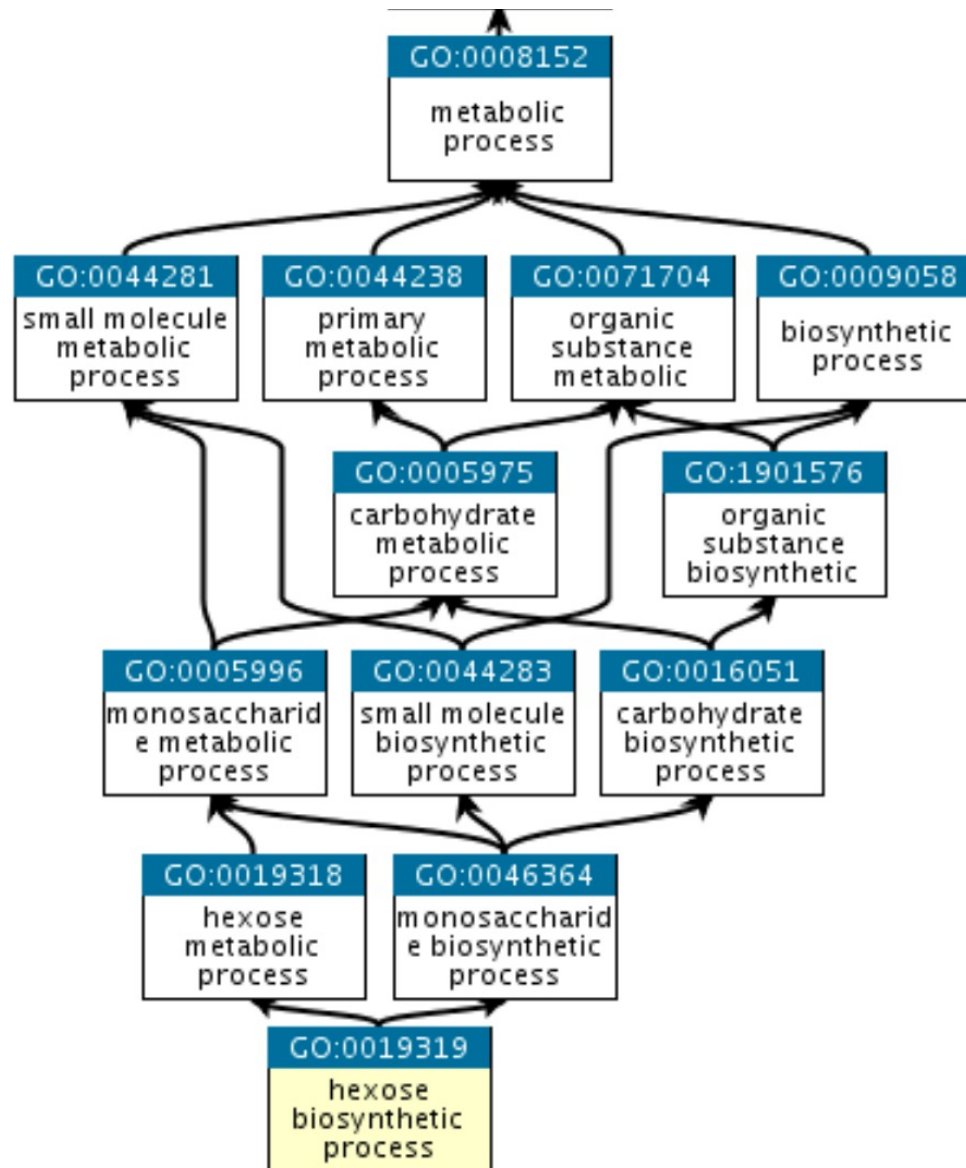
Molecular-level activities performed by gene products. Molecular function terms describe activities that occur at the molecular level, such as [catalysis](#) or [transport](#).

Cellular Component: 4,058 terms

The locations relative to cellular structures in which a gene product performs a function, either cellular compartments (e.g., [mitochondrion](#)), or stable macromolecular complexes of which they are parts (e.g., the [ribosome](#)). Unlike the other aspects of GO, cellular component classes refer not to processes but rather a cellular anatomy.

Gene Ontology (GO)

GO terms are connected as a (loose and non-strict) hierarchy



Testing for enriched GO terms within a list of genes aka gene set enrichment

- You have a list of 200 genes, and 15 are annotated with the GO term tricarboxylic acid cycle.” Is that a little or a lot? What if there are 33 / 7166 total genes with that annotation?
- The overall idea is to apply a statistical test to determine whether certain GO categories are over-represented in a subset of genes relative to the background set of genes (usually the total number of genes in the genome or the total number of genes in your entire dataset)

Fisher's exact test (aka hypergeometric test)

- relies on the hypergeometric distribution
- describes the number of successes in a sequence of n draws from a finite population without replacement
- the classic (Wikipedia) example:
 - if there are 5 green and 45 red marbles in an urn, and you draw 10 times, what is the probability that 4 are green?

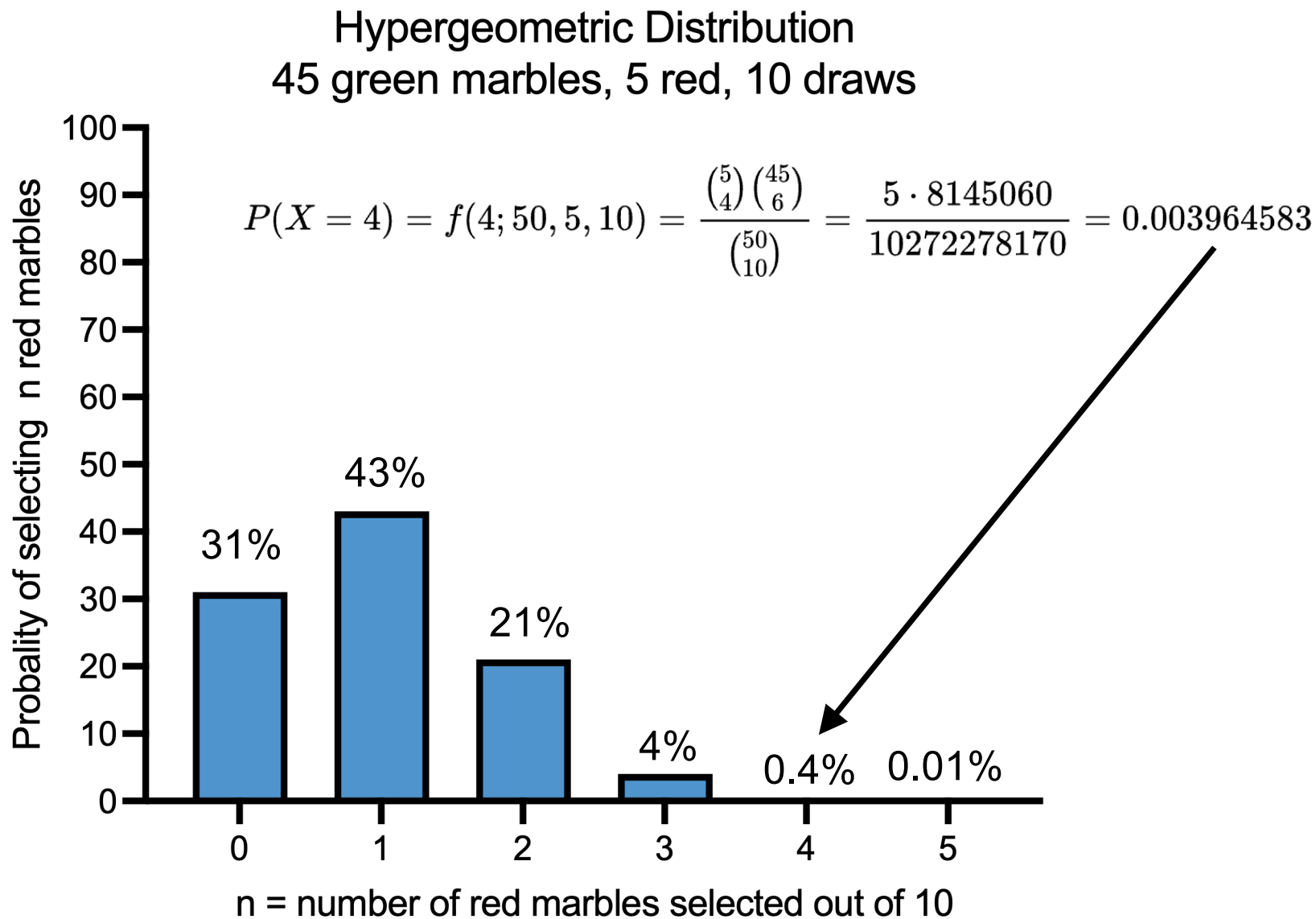
	drawn	not drawn	total
green marbles	k	$K - k$	K
red marbles	$n - k$	$N + k - n - K$	$N - K$
total	n	$N - n$	N

	drawn	not drawn	total
green marbles	$k = 4$	$K - k = 1$	$K = 5$
red marbles	$n - k = 6$	$N + k - n - K = 39$	$N - K = 45$
total	$n = 10$	$N - n = 40$	$N = 50$

$$P(X = k) = f(k; N, K, n) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$$

$$P(X = 4) = f(4; 50, 5, 10) = \frac{\binom{5}{4} \binom{45}{6}}{\binom{50}{10}} = \frac{5 \cdot 8145060}{10272278170} = 0.003964583$$

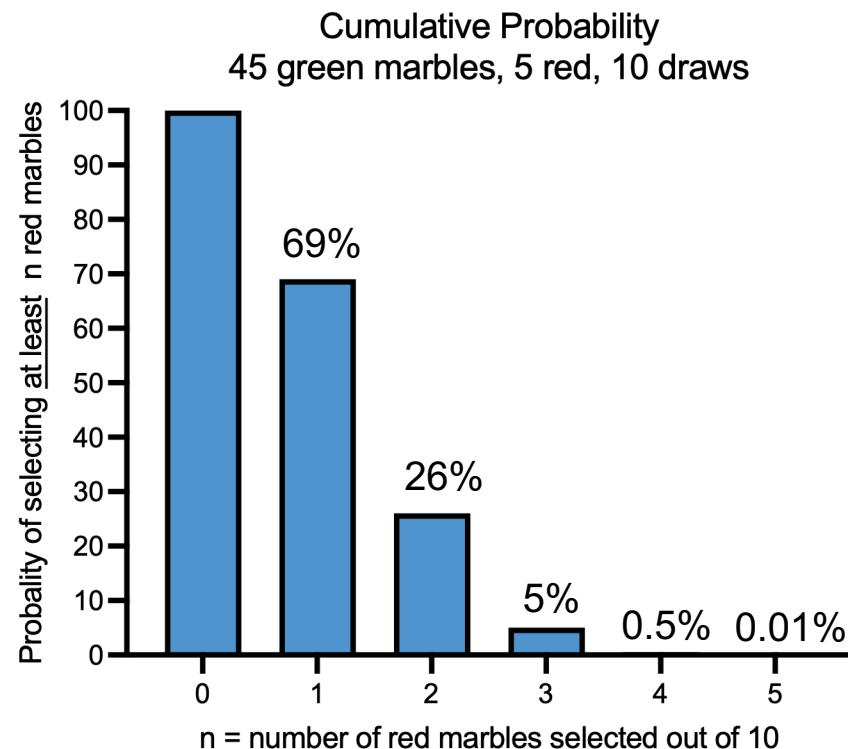
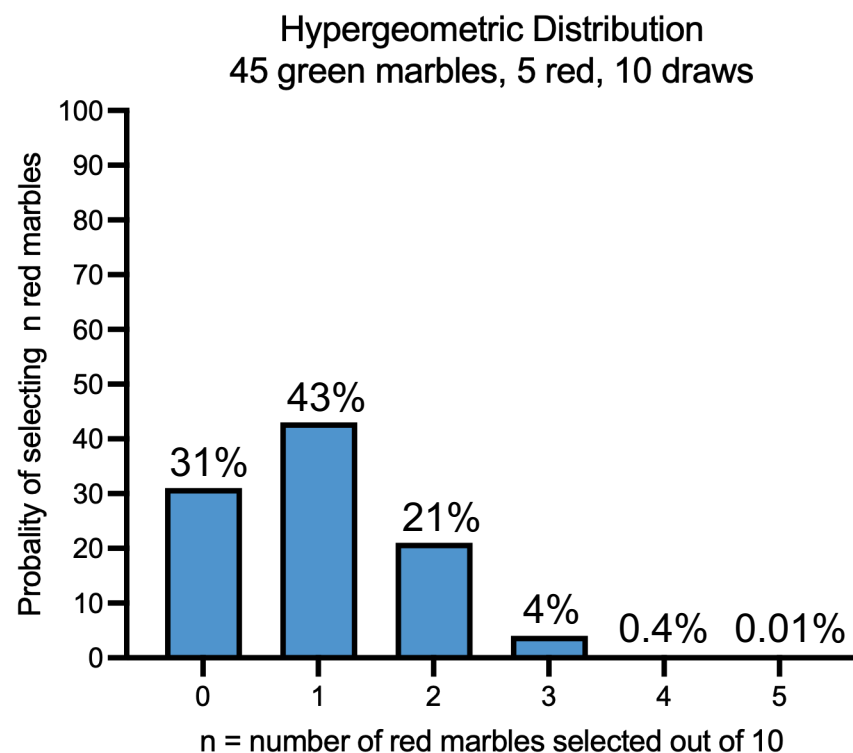
Fisher's exact test (aka hypergeometric test)



Fisher's exact test (aka hypergeometric test)

We usually don't care about the exact probability of drawing a specific value, but instead the cumulative probability of at least that amount

For the urn example, what is the probability of drawing at least 1 or 2 or 3... red marbles?



- <https://shiny.colgate.edu/apps/Collaboratory-Apps/ProbabilityCalculator/>

Fisher's exact test (aka hypergeometric test)

We usually don't care about the exact probability of drawing a specific value, but instead the cumulative probability of at least that amount

Back to our original question:

You have a list of 200 genes, and 15 are annotated with the GO term "tricarboxylic acid cycle." Is that a little or a lot? What if there are 33 / 7166 total genes with that annotation?

Fisher's exact test: the probability (P) of obtaining 15 genes or more with that GO annotation:

1.9×10^{-15}

GO Enrichment Software

In practice, you generally want to input your gene list and then do an automated search for enrichment across all possible GO terms

Gene Ontology term	Cluster frequency	Genome frequency	Uncorrected P-value
ribosome biogenesis	69 of 200 genes, 34.5%	490 of 7166 genes, 6.8%	7.03e-32
rRNA processing	59 of 200 genes, 29.5%	375 of 7166 genes, 5.2%	1.35e-29
rRNA metabolic process	63 of 200 genes, 31.5%	438 of 7166 genes, 6.1%	2.04e-29
ribonucleoprotein complex biogenesis	71 of 200 genes, 35.5%	585 of 7166 genes, 8.2%	1.13e-28
ribosomal large subunit biogenesis	34 of 200 genes, 17.0%	134 of 7166 genes, 1.9%	5.71e-24
ncRNA processing	60 of 200 genes, 30.0%	505 of 7166 genes, 7.0%	2.24e-23
ncRNA metabolic process	66 of 200 genes, 33.0%	638 of 7166 genes, 8.9%	1.98e-22
maturation of LSU-rRNA from tricistronic rRNA transcript (SSU-rRNA, 5.8S rRNA, LSU-rRNA)	20 of 200 genes, 10.0%	54 of 7166 genes, 0.8%	4.44e-18
maturation of LSU-rRNA	22 of 200 genes, 11.0%	70 of 7166 genes, 1.0%	5.36e-18
ribosomal small subunit biogenesis	28 of 200 genes, 14.0%	145 of 7166 genes, 2.0%	1.82e-16
RNA processing	60 of 200 genes, 30.0%	704 of 7166 genes, 9.8%	4.43e-16
tricarboxylic acid cycle	15 of 200 genes, 7.5%	33 of 7166 genes, 0.5%	1.91e-15



<https://go.princeton.edu/cgi-bin/GOTermFinder>

GO Enrichment Software

In practice, you generally want to input your gene list and then do an automated search for enrichment across all possible GO terms

Online Tools:

<https://go.princeton.edu/cgi-bin/GOTermFinder>

<https://pantherdb.org/webservices/go/overrep.jsp>

<https://david.ncifcrf.gov/home.jsp>

R Tools:

clusterProfiler (which we'll use)

gprofiler2

topGO

GO Enrichment Software

all packages include a choice to account for multi-testing (since you are searching for enrichment against hundreds to thousands of GO terms):

Two main methods:

- 1) false discovery rate (FDR): set the acceptable percentage of false positive GO term enrichments (usually 5% or 1/20 enriched categories)
- 2) Bonferroni correction: multiplies the p -values by the number of tests

Bonferroni is very conservative (if are performing 100 tests, you need an unadjusted p -value of 0.0001 to reach an adjusted $p = 0.01$)

Both Bonferroni and FDR assume independent tests, which is not true for GO terms (because they are a hierarchy)

Look at the methods used for multiple-test correction—for example, some modifications include collapsing related GO terms to reduce the Bonferroni “penalty”, while the elim algorithm of TopGo addresses dependency of nested terms

GO Enrichment Software

E.g., the Princeton GO Term Finder can provide both Bonferroni-corrected p -values and the FDR: our unadjusted p -value was 1.9×10^{-15}

Gene Ontology term	Cluster frequency	Genome frequency	Corrected P-value	FDR	False Positives
ribosome biogenesis	69 of 200 genes, 34.5%	490 of 7166 genes, 6.8%	3.37e-29	0.00%	0.00
rRNA processing	59 of 200 genes, 29.5%	375 of 7166 genes, 5.2%	6.49e-27	0.00%	0.00
rRNA metabolic process	63 of 200 genes, 31.5%	438 of 7166 genes, 6.1%	9.79e-27	0.00%	0.00
ribonucleoprotein complex biogenesis	71 of 200 genes, 35.5%	585 of 7166 genes, 8.2%	5.44e-26	0.00%	0.00
ribosomal large subunit biogenesis	34 of 200 genes, 17.0%	134 of 7166 genes, 1.9%	2.74e-21	0.00%	0.00
ncRNA processing	60 of 200 genes, 30.0%	505 of 7166 genes, 7.0%	1.07e-20	0.00%	0.00
ncRNA metabolic process	66 of 200 genes, 33.0%	638 of 7166 genes, 8.9%	9.52e-20	0.00%	0.00
maturation of LSU-rRNA from tricistronic rRNA transcript (SSU-rRNA, 5.8S rRNA, LSU-rRNA)	20 of 200 genes, 10.0%	54 of 7166 genes, 0.8%	2.13e-15	0.00%	0.00
maturation of LSU-rRNA	22 of 200 genes, 11.0%	70 of 7166 genes, 1.0%	2.57e-15	0.00%	0.00
RNA processing	60 of 200 genes, 30.0%	704 of 7166 genes, 9.8%	2.13e-13	0.00%	0.00
ribosomal small subunit biogenesis	27 of 200 genes, 13.5%	145 of 7166 genes, 2.0%	8.28e-13	0.00%	0.00
tricarboxylic acid cycle	15 of 200 genes, 7.5%	33 of 7166 genes, 0.5%	9.20e-13	0.00%	0.00

